TRƯỜNG ĐẠI HỌC BÁCH KHOA HÀ NỘI **TRƯỜNG ĐIỆN – ĐIỆN TỬ** **Đề số: 01** **Tổng số trang: 4**	**ĐỀ THI CUỐI KỲ 2021.2** **Học phần: ET3116 – KT VI XỬ LÝ** **Ngày thi: 03/08/2022** **Thời gian làm bài: 75 phút** *(Được sử dụng tài liệu. Nộp đề thi cùng với bài làm)*
Ký duyệt Người ra đề:	Trưởng Nhóm chuyên môn phụ trách HP: Nguyễn Quốc Cường
Student name:	**Student ID:**
Total points: 290	**Get: /290**

Note that:

- For questions with required explanation, the right choice(s) of your answer is only made 40% of points. The rest of 60% of points is made by details you provide.

PART A: QUESTIONS FOR THE INTEL MICROPROCESSORS				
Question 1. What is the right instruction to move data from a memory location to a register?			**Point: 20**	**Get:**
A. MOV [1234h], AL	B. MOV [SI+1234h], AL	C. MOV AL, 1234h	D. MOV AL, [SI]	
For the choice, please give brief explanation for the data size in bits, and what is the address of memory location? …… ……				
Question 2. What is the right instruction to move data from a register to a memory location?			**Point: 20**	**Get:**
A. MOV [1234h], AH	B. MOV [SI+1234h], AX	C. MOV AH, 1234h	D. MOV AX, [SI]	
For the choice, please give brief explanation the data size(s) in each case. …… ……				

Question 3. Please indicate the memory segment(s) in which a microprocessor is working while executing the below instructions, and what is data size/range in each instruction (if any)?	**Point: 50**	**Get:**

No.	**Instruction**	**Segment**
1.	MOV AL, BH	

2.	MOV AX, [1234h]	
3.	PUSH AX	
4.	POP BX	
5.	JMP [BX]	
6.	JNZ LabelA	
7.	MOVSB	
8.	ROR CL,4	
9.	IN AL, 0Fh	
10.	OUT DX, AL	

Question 4. Which is the register(s) with the content being changed in below instructions?	**Point: 50**	**Get:**

No.	**Instruction**	**Affected register**	**Explanation for activity of microprocessor**
1.	ADD AH, CH		
2.	SUB DL, CL		
3.	LOOP LabelA		
4.	DIV BH		
5.	MUL BL		
6.	STI		
7.	PUSH AX		
8.	MOV AH,01h INT 21h		
9.	IN AL, DX		
10.	OUT 0Fh, AX		

Question 5. How many schemes can be used to exchange data between an I/O device and memory?	**Point: 10**	**Get:**

A. 1	B. 2	C. 3	D. other is filled……

Please give details about them.

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Question 6. What is the content of AL after the below piece of code being executed? STC MOV AL,0Fh RCL AL,4			**Point: 20**	**Get:**
A. F0H	B. 7CH	C. F8h	D. 3Eh	
Please detail in step-by-step. …… …… ……				

Question 7. What is the content of AL after the below piece of code being executed? MOV CL,8 MOV AX,0F LabelA: DEC AX LOOP LabelA			**Point: 20**	**Get:**
A. 07h	B. 08h	C. 09h	D. 0Ah	
Please give details for your solution. …… …… ……				

Question 8. What is the content of interrupt vector table?			**Point: 10**	**Get:**
A. Data for programs	B. Stack for programs	C. Code for programs	D. Addresses of interrupt functions	

Question 9. What is the difference between RET and IRET?			**Point: 20**	**Get:**
A. Nothing	B. In the operation of microprocessor in ending a function	C. In the operation of microprocessor in beginning a function	D. All of them	
Please give the brief explanation for each of your choice(s). …… ……				

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Question 10. The function named PAC can be called by which of below instructions?			**Point: 20**	**Get:**
A. CALL PAC	B. MOV BX, OFFSET PAC CALL [BX]	C. MOV [1234h], OFFSET PAC CALL [1234h]	D. All of them	
Please give the brief explanation for each of your choice(s). …… …… …… ……				

PART B: QUESTIONS FOR THE ARM MICROPROCESSOR (CORTEX-M)				
Question 11. How many general-purpose 32-bit <u>visible</u> registers of ARM?			**Point: 20**	**Get:**
A. 13	B. 14	C. 15	D. 16	
Please list their name. ……				
Question 12. What is the classification of ARM microprocessor belonging to?			**Point: 10**	**Get:**
A. CISC	B. RISC	C. Harvard	D. Von Neumann	
Question 13. How many register modes in ARM microprocessor?			**Point: 20**	**Get:**
A. 5	B. 6	C. 7	D. 8	
Please list them down: …… ……				